Does Simulation Add Information Beyond Persistence? A Financial Digital Twin Tested on a Survivorship-Free Panel of SEC Filers

Spilios Dimakopoulos*

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Abstract

Financial digital twins simulate how a company's cash, debt and margins respond to a shock. We ask whether that simulation adds information that simple persistence rules do not. FinTwin is a system-dynamics model calibrated for each company from point-in-time SEC XBRL filings; we test it on a survivorship-free universe of 5,220 non-financial filers (21,531 firm-years, 2014–2024) with a census of 419 bankruptcy petitions. The main test feeds each twin the revenue that actually followed and predicts eight quarters of operating cash flow. On its own the twin is the least accurate of eight models: its median error is 11.3 percentage points of revenue, against 6.9 for a 16-quarter cash-flow margin. A statistic from the first version of this paper, the correlation of the twin's departures from a four-quarter margin with the realized ones (0.32), is reproduced by the 16-quarter margin alone (0.39): it measured mean reversion, not transmission. The twin's departures do carry information that none of six simple rules has: their correlation with the residual of the 16-quarter margin is 0.22 (95% CI 0.19–0.24), and they survive an encompassing regression ($t = 9.9$). Out of sample they are worth little: adding 0.10–0.15 of them to the best rule lowers its error by 3.3%, but adding a simple cash-revenue regression the same way does almost as well; only for free cash flow does the twin improve on every simple rule in such a combination, by 0.22 percentage points. The twin over-reacts: its cash flow responds to revenue 2.3 times as strongly as firms' does, because its prior on fixed costs is too high; a prior selected walk-forward lowers its median error from 11.7 to 9.3 points, still above the margin's. In ranking firms, simulation adds nothing: the twin's score predicts bankruptcy within two years with an AUC of 0.85, its scoring formula applied to the opening balance sheet reaches 0.87, and among already-weak firms the two are level. Given each firm's industry shock, however, the twin's fragility predicts which firms burned more cash than usual in the 2015–16, 2020 and 2022 downturns. Filing-based simulation mostly encodes persistence; its marginal value is small and confined to translating a known shock into free cash flow.

Keywords: digital twin; system dynamics; corporate liquidity; cash-flow prediction; bankruptcy prediction; stress testing; XBRL; forecast encompassing.
JEL: G32, G33, G17, C53, C63, M41.

*Independent researcher. E-mail: spiliosdimakk@gmail.com. Code and data: https://github.com/SpiliosDimakopoulos/fintwin, archived at doi:10.5281/zenodo.23434606. I thank the researchers who commented on the first version; the null tests, the census of bankruptcies and the survivorship-free sample in this version respond to their comments.

1 Introduction

A corporate stress test asks a conditional question: *if* revenue falls by a third for a year and debt markets close, does the company run out of cash, and when? Credit analysts usually answer it with ratios — leverage, coverage, liquidity — or with accounting distress scores (Beaver, 1966; Altman, 1968; Ohlson, 1980; Zmijewski, 1984). A simulation model promises more: it represents the mechanisms through which a shock becomes a cash shortfall (fixed costs, working capital, maturing debt, credit lines, the loss of market access) and runs them forward. System dynamics provides the language for such models (Forrester, 1961; Sterman, 2000), and engineering calls a simulation that is kept synchronized with its physical counterpart a digital twin (Grieves and Vickers, 2017; Tao et al., 2019; Kritzinger et al., 2018).

The question this paper asks is whether the simulation adds information. A model that runs a stock-and-flow structure over thousands of Monte Carlo paths is expensive to build and to explain; if a trailing cash-flow margin, a regression of cash on revenue or an accounting score computed from the same filings predicts as well, the structure is ornament. The first version of this paper reported that it did add information: fed the revenue that actually followed, the twin's departures from a four-quarter margin rule correlated with the realized departures at 0.30, positive in every year. Referees pointed out that the statistic rewards any predictor that is smoother than the rule, because margins mean-revert (Fama and French, 2000) and the twin's costs are estimated over 16 quarters. They were right: a 16-quarter margin, with no simulation at all, scores 0.39 on the same statistic.

This version therefore tests the twin against the rivals that matter. Section 2 sets out how the twin relates to distress scores, structural credit models, financial-planning models and top-down stress tests, and states three hypotheses. The data are every non-financial SEC filer with at least $50 million of revenue, including the ones that later disappeared, and every bankruptcy petition they filed (Section 3). The primary test (Section 5) is a single one: given the realized revenue path, predict cumulative operating cash flow over eight quarters, and compare the twin with six simple rules that see the same revenue — margins over 4, 8 and 16 quarters, an industry-shrunk margin, and per-firm regressions of cash flow on revenue with and without working-capital timing. The decision rule for accepting the transmission claim was fixed and committed to the repository before the test was run.

The results are mixed in a way that is informative. In levels the twin is the worst of the eight models (Section 6): its fixed-cost structure makes it overreact to revenue changes. Its departures from the rules nevertheless contain information that none of the rules has, in a regression sense. Out of sample that information is worth little: a forecast that adds a small fraction of the departures to the 16-quarter margin is about 3.3% more accurate than the margin alone, but most of that gain is the familiar benefit of combining two forecasts — a simple regression rule added in the same way achieves almost the same — and only for free cash flow does the twin's contribution exceed that of every simple rule. In ranking firms by distress (Section 8), the twin's score is as good as the best accounting score and no better than its own scoring formula evaluated without simulation, including in the hard task of telling which already-weak firms will file. The feedback loops do not change rankings but are the only channel through which the model produces cash depletion and new borrowing, and they improve the prediction of those liquidity events slightly (Section 7). The 2020 shock is one episode among eleven; it shows that, once each firm is given its industry's realized revenue shock, fragility predicts the abnormal cash burn that a uniform scenario

could not (Section 9). Probabilistic forecasts are well calibrated on average after conformal rescaling but not in the stress year (Section 10), and the main results are insensitive to the model's assumed parameters except for the prior on fixed costs (Section 11).

The paper makes three contributions. It shows how easily a simulation model's apparent skill can come from persistence, and provides a ladder of simple rules, an encompassing test (Chong and Hendry, 1986; Fair and Shiller, 1990) and an out-of-sample combination (Bates and Granger, 1969) that separate the two. It provides an open, survivorship-free, point-in-time panel of all non-financial XBRL filers with a census of bankruptcy petitions, on which fundamental models can be tested without hindsight. And it gives a measured answer to what a filing-based financial twin is for: scenario analysis of a specified shock, with its output shrunk toward persistence, not early warning.

2 Conceptual framework

2.1 Twin, shadow or simulation model?

The term *digital twin* is used loosely, so we state which of its defining properties the model has. Kritzinger et al. (2018) separate three levels by the direction and automation of data flows between a physical system and its virtual counterpart: a *digital model* is updated by hand, a *digital shadow* is updated automatically from the physical system, and a *digital twin* also feeds decisions back into the physical system automatically. By this taxonomy FinTwin is a digital shadow with a human in the loop. It is synchronized automatically: every week a scheduled job reads each company's newest filings, re-estimates every parameter and re-runs the simulations, and its state can be rewound to any past filing date. It represents the firm's financial system as stocks and flows that map one-to-one onto the balance sheet, rather than as a statistical summary of it. And it supports decisions through what-if analysis: scenarios, management levers (cutting capital expenditure or payouts, drawing the revolver, issuing debt or equity) and reverse stress tests. It does not close the loop automatically, since no decision taken in the model is executed in the firm, and its synchronization frequency is bounded by the quarterly reporting cycle. We keep the term because it is established in the system-dynamics and engineering literature for models with these properties, and because organizational twins are increasingly discussed in management (Lyytinen et al., 2024), but the evidence in this paper concerns the model, not the label.

2.2 Relation to existing models

Table 1 places the twin among four families of models that address related questions.

Accounting-based distress scores (Beaver, 1966; Altman, 1968; Ohlson, 1980; Zmijewski, 1984) and their hazard-model successors (Shumway, 2001; Chava and Jarrow, 2004; Campbell et al., 2008) map ratios to the probability of an event. They are estimated on default samples and are not conditional on a scenario. *Structural credit models* (Merton, 1974; Leland, 1994) derive default probabilities from asset value and volatility, inferred from equity prices; their state variable is a single asset value, not cash, working capital or a maturity ladder, so they cannot say whether a given revenue path exhausts liquidity before the debt is due. *Corporate financial-planning models* (Carleton, 1970; Warren and Shelton, 1971) are simultaneous-equation projections of the financial statements under managerial assumptions; they are close in spirit to the twin's accounting core but are deterministic, specified by the analyst and rarely validated out of sample. *Top-down corporate*

Table 1: Models of corporate financial fragility

	Distress scores	Structural credit	Financial planning	Top-down stress tests	FinTwin
Question	probability of default	default probability from market values	projected statements	sector shock → firm losses	shock path → cash, debt, breaches
Conditional on a scenario	no	partly	yes	yes	yes
State variables	ratios	asset value	statements	balance sheet	9 balance-sheet stocks
Liquidity timing (maturities, credit lines)	no	no	sometimes	sometimes	yes
Feedback (spreads, market access, management response)	no	endogenous default	no	rarely	4 loops
Uncertainty	event probability	diffusion	none	scenarios	Monte Carlo
Parameters	pooled estimates	market-implied	analyst	mixed	per firm, from filings
Updating	periodic re-estimation	daily	manual	annual	weekly, automatic

stress tests run by central banks and the IMF (Caceres et al., 2020) apply macroeconomic and sector-level scenarios to firm-level data and project income, funding needs and debt service under common assumptions, without firm-specific feedback between financing conditions and operations. Finally, the accounting literature on *cash-flow prediction* (Dechow et al., 1998; Barth et al., 2001) and on the mean reversion of profitability (Fama and French, 2000) provides the simplest rivals for the twin's transmission mechanics: rules that map revenue or earnings into cash through margins, accruals and working capital.

The twin combines elements that these families keep apart: an accounting identity that holds on every path, firm-specific parameters estimated automatically from filings, contractual debt maturities and credit lines, path-dependent financing constraints (feedback loops), and a probabilistic description of uncertainty. Whether this combination predicts anything that the simpler members of each family do not is an empirical question. The rest of the paper answers it against the strongest simple rival in each case.

2.3 Further related work

Corporate liquidity. Firms hold cash and credit lines to insure against cash-flow shortfalls (Opler et al., 1999; Almeida et al., 2004; Bates et al., 2009; Sufi, 2009). The 2020 shock offered a natural experiment. Fahlenbrach et al. (2021) show that firms with more financial flexibility before the pandemic suffered smaller stock-price declines, Ramelli and Wagner (2020) and Ding et al. (2021) document that pre-pandemic leverage and cash holdings shaped stock returns, and Acharya and Steffen (2020) describe the dash for cash through credit-line drawdowns and bond issuance by firms at risk of downgrade. These studies explain market reactions with balance-sheet characteristics. We use the same episode to evaluate a model that was not estimated on it, and we measure outcomes in cash rather than in prices.

Forecast evaluation. We evaluate the twin's predictive distributions with the continuous ranked probability score and probability integral transform histograms (Gneiting and Raftery, 2007; Gneiting et al., 2007), compare predictive accuracy with Diebold–Mariano tests (Diebold and Mariano,

1995) using clustered standard errors (Petersen, 2009; Cameron et al., 2008), benchmark against exponential smoothing in state-space form (Hyndman et al., 2008), and recalibrate intervals with online split-conformal prediction (Vovk et al., 2005; Gibbs and Candès, 2021).

2.4 Hypotheses

The design of Section 5 tests three hypotheses, each against rivals that see the same information.

H1 (ranking). Under a common scenario, the twin's stress score ranks firms' subsequent distress better than accounting scores and simple cash-flow benchmarks computed from the same filings.

H2 (transmission). Given the revenue path that actually occurred, the twin's cash-flow response contains information about realized cash flows that is not contained in margin rules (including long-window and industry-shrunk margins) or in a per-firm regression of cash flow on revenue and its change.

H3 (feedback). The feedback loops improve the prediction of the outcomes they are designed to produce — new borrowing, cash depletion, dividend and capital-expenditure cuts — not only the cross-sectional ranking of firms.

A formal decision rule was fixed before the full-sample run for H2, the primary hypothesis; H1 and H3 are evaluated descriptively. H2 is accepted only if the twin's incremental correlation is positive against every simple rule with a confidence interval that excludes zero and its coefficient in an encompassing regression (Chong and Hendry, 1986; Fair and Shiller, 1990) is significant at 5%.

3 Data

3.1 Point-in-time XBRL panels

The SEC `companyfacts` endpoint returns every value a registrant has tagged in XBRL since 2009, with the period it describes, the form, the filing date and the accession number. We build a quarterly panel of 38 fields per company from 72 tags (income statement, cash-flow statement, balance sheet, debt by component, impairments, customer deposits and committed credit facilities) together with the contractual debt-maturity schedule. Four problems make raw XBRL unusable for backtests, and each is handled explicitly.

Tag drift. Companies change tags over time (for example from `SalesRevenueNet` to the revenue-from-contracts tag introduced with ASC 606). Each field has an ordered list of fallback tags, merged period by period. Total debt is assembled from its current, non-current, commercial-paper and short-term components; operating income is rebuilt as pre-tax income plus interest when it is not tagged.

Fiscal calendars and year-to-date reporting. Cash-flow statements in 10-Q filings are cumulative, and the fourth quarter appears only inside the 10-K, so facts that share a fiscal-year start are differenced (H1−Q1, 9M−H1, FY−9M). Retailers on 52/53-week calendars report one quarter of 16 or 17 weeks each year (the first quarter at Kroger, the fourth at Costco and AutoZone). The first version of the pipeline accepted only periods of 75 to 105 days and silently dropped that quarter, so trailing-twelve-month sums for these companies covered three fiscal quarters. A quarter is now any period of 75 to 126 days.Across the 5,219 companies, gaps longer than 126 days in the

quarterly spine fell from 1,043 (in 562 companies) to 748 (530), and the panels of 36 companies became complete. Of the remaining gaps, 0 fall where the company filed no 10-Q or 10-K at all (late filers, transition periods, deregistration) and 0 where a filing exists but tags no revenue, often subsidiaries and limited-liability companies that file for their debt. The last quarter in each panel matches the company's last 10-Q or 10-K in 87% of cases, after the fallback described next.

Missing filings. The `companyfacts` database does not contain every XBRL filing: in October 2026 it still lacked the July 10-Qs of HCA Healthcare and Coca-Cola. For every 10-K or 10-Q that the company's filing index lists but the database lacks, the loader reads the filing's own XBRL instance and adds its entity-wide facts with the filing's accession number and date, so point-in-time loading treats them like any other fact. Industry-specific revenue tags (utilities, oil and gas, real estate, contractors, services) are used as a last resort for periods that no general revenue tag covers.

Impairments. Goodwill, intangible and long-lived-asset impairments are added back to operating income, as credit analysts do, because they are non-cash and would otherwise dominate trailing coverage ratios.

Vintages. For a cutoff τ, the loader keeps only facts *filed* on or before τ and, for each period, the latest of those filings, so later restatements are excluded. Every model input, benchmark and calibrated parameter is computed from vintages; realized outcomes are measured on the latest data. A company enters a cutoff only if it has 16 quarters of history and its last known quarter ended at most 200 days before τ, which excludes companies that had stopped filing.

Short-term rates, the term spread and the Baa spread come from FRED; none of these series is revised. Nominal GDP, which is revised, is taken from the vintage published on the cutoff date in the Real-Time Data Set for Macroeconomists (Croushore and Stark, 2001). NBER recession dates since 1947 drive the regime model.

3.2 A survivorship-free universe

The first version of the paper used the current S&P 500, which conditions every backtest on survival to 2026. The universe is now built from EDGAR itself. For each calendar year from 2012 to 2024, the XBRL `frames` API returns one revenue fact for every company that reported one, including companies that were later acquired, delisted or liquidated. A company enters the universe if it reported at least $50 million of revenue in some year under any revenue tag the loader uses and is not a financial company or REIT (SIC 6000–6799; managed-care insurers, which GICS classifies as health care, are kept). This gives 5,220 companies. At each cutoff, a company is in the sample if its own filings on that date meet the history requirement and its trailing revenue is at least $50 million. Industries are the Fama–French twelve (Fama and French, 1997).

3.3 A census of bankruptcies

Item 1.03 of Form 8-K ("Bankruptcy or Receivership") reports the appointment of a receiver or a bankruptcy proceeding of the registrant, and registrants also use it when a reorganization plan is confirmed. Reading every company's complete filing history from the EDGAR `submissions` API, 463 companies in the universe filed at least one such 8-K; 63% filed more than one, mostly a

petition followed by a confirmation. The census recovers 33 of the 35 Chapter 11 filers of 2020 in the convenience sample of the first version; the missing ones (CEC Entertainment and Hornbeck Offshore Services) did not file an Item 1.03 8-K. Firms that stopped filing before petitioning are not counted, and a few 1.03 filings concern subsidiaries or operating reports, so the event is measured with some error. A firm-year has a bankruptcy event if a petition follows the cutoff within eight quarters; firm-years with a petition in the two preceding years are excluded, because the company is already in or just out of bankruptcy. This replaces the convenience sample of 27 prominent 2020 filers used before.

3.4 The S&P 500 as of 20 February 2020

For comparison with the first version, the 2020 analysis is repeated on the index as it stood on 20 February 2020, taken from the Wikipedia constituents table of that day (which lists CIKs) and checked against the independent membership history of `github.com/fja05680/sp500`; the two lists agree on all 505 tickers.

4 The twin

4.1 Stocks, flows and feedback loops

The model is quarterly and vectorized over Monte Carlo paths. Its stocks are cash, receivables, inventory, payables, customer deposits, net property, plant and equipment, term debt, revolver drawings and equity; a residual "other net assets" stock makes the opening balance sheet balance. The accounting identity assets − liabilities − equity = 0 holds on every path and in every quarter, which is verified by property-based tests over random scenarios.

Demand. Deseasonalized log revenue growth follows an AR(1) with Student-t innovations and a two-state Markov recession regime s_t (Hamilton, 1989):

$$g_t = \mu + \phi\,(g_{t-1} - \mu) + \sigma_g(1 + 0.5\,s_t)\varepsilon_t^d + \delta\,s_t, \qquad \mathrm{Rev}_t = B_{t-1}e^{g_t}\,S_{q(t)}\,(1 + \xi_t), \tag{1}$$

where S_q are seasonal factors, ξ_t is the scenario's revenue shock, the regime's entry and exit probabilities are the NBER base rates, and $\delta = -\beta\,\Delta_{\mathrm{GDP}}$ scales the average nominal-GDP shortfall in recessions by the company's estimated revenue cyclicality β.

Costs and working capital. Cash operating costs are $F(1+\pi)^t c_t + (v + m_t + \kappa_t)\,\mathrm{Rev}_t$, with fixed costs F, a variable-cost ratio v, a mean-reverting margin shock m_t correlated with demand, a scenario cost shock κ_t and a restructuring multiplier c_t. Receivables and inventory adjust to days-based targets with first-order delays (collections and production lag demand); payables and customer deposits adjust immediately, so a collapse in bookings triggers deposit refunds.

Financing. Debt matures according to the company's own XBRL maturity schedule. A share of maturities is refinanced according to the company's revealed rollover policy, at a rate equal to the short rate plus a term premium and a rating-dependent spread. The treasury raises new debt when cash falls below a target, draws on its committed revolver below a revealed minimum cash

balance, and sweeps idle cash above a cap into debt repayment. A *liquidity breach* occurs when cash is negative after all of these.

Feedback loops. Four loops govern the response to stress (Figure 1):

- **R1, leverage–spread spiral.** Trailing interest coverage maps to a synthetic rating and credit spread (Damodaran, 2026); refinanced and new debt carry that spread, which raises interest expense and lowers coverage further.
- **R2, market-access cliff.** The share of a funding need that can be placed falls from 100% (coverage at least 1.5×) to 60% and 30% as coverage deteriorates, and to zero while the scenario closes debt markets. Unfunded maturities drain cash.
- **B1, cash conservation.** When cash approaches its floor or coverage falls below 2.5×, dividends and buybacks stop and capital expenditure falls to maintenance.
- **B2, delayed restructuring.** After two quarters below the target margin, fixed costs fall by 4% per quarter to a floor of −15%, and recover when margins do.

R1 and R2 are reinforcing: they turn a revenue shock into a financing shock. B1 and B2 are balancing: they are management's response. Whether a shock fades or spirals depends on which loops dominate, which is the reason for building the model as a system-dynamics model (Forrester, 1961; Sterman, 2000; Morecroft, 2015) rather than as a set of ratios.

Health score. Each path is summarized by a 0–100 health score that combines liquidity runway (30%), the synthetic rating (30%), net debt/EBITDA (20%) and Altman's Z″ (Altman and Hotchkiss, 2006) (20%), capped at 25 after a breach. A company's *health score* under a scenario is the median, across Monte Carlo paths, of each path's average score over the horizon, and its *breach probability* is the share of paths with a liquidity breach by the end of the horizon. The *stress health score* uses the demand-freeze scenario and an eight-quarter horizon. The product version of the score also scores coverage and leverage against sector norms; because those norms are estimated from current data, every result in this paper uses the unadjusted score unless stated otherwise.

4.2 Calibration

All parameters are estimated from the company's own point-in-time history (Appendix Table 16). Seasonal factors are median ratios to a centred moving average; trend growth is the median of the last 12 quarters shrunk halfway toward 1.2% per quarter; volatilities are robust (median absolute deviation) AR(1) residual standard deviations, with Student-t tails fitted by maximum likelihood. The fixed/variable split comes from a penalized regression of cash operating costs on revenue over 16 quarters, shrunk toward a 35% fixed share when revenue has not varied enough to identify it. The rollover share is inferred from the two-year debt trend, and the minimum operating cash balance from revealed behaviour (half the 10th-percentile cash balance over three years). A Vasicek short rate (Vasicek, 1977) is fitted to the 3-month Treasury bill. Parameters that are assumptions rather than estimates (for example a credit facility equal to 8% of revenue when none is tagged) are labelled as such in the software's output.

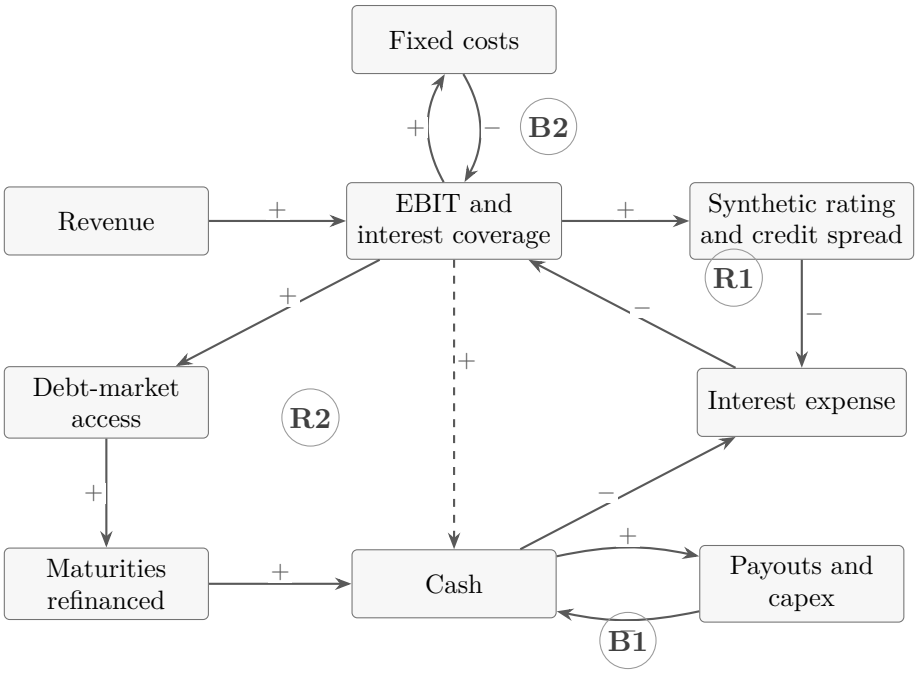


Figure 1: The four feedback loops in causal-loop notation. A "+" ("−") link means that the variables move in the same (opposite) direction; a loop is reinforcing (R) when the product of its signs is positive and balancing (B) when it is negative. R1, leverage–spread spiral: lower coverage worsens the rating, which raises interest and lowers coverage further. R2, market-access cliff: lower coverage reduces the share of maturities that can be refinanced, so cash falls, revolver drawings raise interest, and coverage falls further. B1, cash conservation: payouts and capex are cut as cash falls. B2, delayed restructuring: fixed costs are cut as margins fall. The dashed link (operating cash flow) is not part of a loop. The full model also contains working capital, customer deposits, the debt-maturity ladder, taxes and the recession regime.

4.3 Scenarios

Eight presets (rates +200 bp, sales −20%, recession, demand freeze, stagflation, credit crunch, supply-chain shock and a baseline) are expanded into per-quarter shock paths. The *demand freeze*, used throughout the evaluation, is a generic 2020-style shutdown: revenue −70% at the trough for three quarters with an eight-quarter recovery, credit spreads +400 bp, debt markets closed for two quarters and policy rates cut by 150 bp. The same scenario is applied to every company, so a low stress score measures structural fragility to a common shock, not a forecast of exposure. Reverse stress testing finds the smallest shock that produces a 50% breach probability, and Sobol indices (Sobol', 2001; Saltelli et al., 2010) decompose the variance of the health score over the joint scenario space.

5 Evaluation design

5.1 The primary test: transmission

For every firm and every cutoff (15 February 2014–2024), the twin is calibrated on the filings available at the cutoff. The revenue the company actually reported over the eight following quarters then replaces the twin's simulated revenue, 500 paths are run, and the median path of quarterly

operating cash flow (and free cash flow) is taken. The outcome is cumulative operating cash flow over the eight quarters as a percentage of revenue in the four quarters before the cutoff. This isolates transmission — how a given revenue path becomes cash — from the scenario, which no filing can predict. Every rival sees the same realized revenue, quarter by quarter:

N1 the trailing four-quarter cash-flow margin times revenue (the rule used in the first version);

N2 the same over eight quarters, and

N3 over sixteen quarters, the same window as the twin's cost regression;

N4 the four-quarter margin shrunk toward the leave-one-out median margin of the firm's industry in the same year, with the weight chosen each year to minimize errors on earlier years only;

N5 a per-firm regression of quarterly cash flow on revenue over 16 quarters (a fixed and a variable part: operating leverage without simulation);

N6 the same regression with the change in revenue added (working-capital timing).

T0 is the twin with its four feedback loops switched off and T1 the full twin. N5 and N6 are the "simpler dynamic financial models" a referee asked about, and N6 also answers the point that the twin, unlike a margin rule, sees the timing of revenue: so does N6.

We report four statistics. *Errors in levels*: median and mean absolute error (outcomes winsorized at the 1st and 99th percentiles for the mean). *Incremental correlation*: the Spearman correlation between a model's departure from a base rule and the realized departure from it, against every base, not only the weakest. *Encompassing*: an OLS regression of the realized departure from N1 on the departures of the twin, N3, N4 and N6, with year effects and firm-clustered errors; a significant twin coefficient means its forecast contains information the others do not (Chong and Hendry, 1986; Fair and Shiller, 1990). *Out-of-sample combination*: for test year Y, the prediction $\mathrm{N}k + \lambda(\mathrm{T1} - \mathrm{N}k)$ with λ chosen on years before Y (Bates and Granger, 1969), compared with Nk alone and with the same combination built from a simple rule.

Decision rule. H2 is accepted only if the twin's incremental correlation against each of N3, N4 and N6 has a 95% confidence interval above zero and its encompassing coefficient is significant at 5%. The rule was committed to the repository before the full-sample run (commit `460cdda`). An earlier draft of the rule also required the best simple rule to reach less than 80% of the twin's statistic against N1; it was replaced before the run because a synthetic test showed it rejects a twin with genuine information whenever the four-quarter rule is very noisy. That draft was never committed, so only the final rule can be verified in the repository history.

Second round. A second set of tests — a revised specification, liquidity-event timing and thresholds, other downturns, the mechanism of the free-cash-flow result, group-conditional and adaptive calibration, two-way clustering and small firms — was added after the first results, with its decision rules committed before it was run (commit `7a5d9b1`). One implementation error found after that run, an adaptive-conformal update applied per forecast instead of per batch of yearly forecasts, was corrected; the rule itself was not changed.

Specifications added after the run. Three analyses were added after the first full-sample results and are exploratory: the combinations built on the 16-quarter margin (the pre-specified combination used the four-quarter margin; the 16-quarter margin was chosen as a base because it turned out to be the most accurate rule), the unconstrained encompassing regression in levels, and

the combinations of the 16-quarter margin with each simple rule, which control for the gains that any combination of forecasts delivers (Bates and Granger, 1969; Smith and Wallis, 2009).

5.2 Secondary tests

Liquidity decisions. The same runs give the share of paths on which cash falls below half its opening level, net new borrowing exceeds 5% of revenue, dividends are cut by half or capital expenditure by a quarter. They are compared with the realized events and with a static runway calculation that uses the same inputs: the trailing free-cash-flow margin on the realized revenue, minus the current payout run-rate and the maturities the company does not usually refinance. Each loop is switched off in turn.

Ranking. Without the realized revenue, the twin's baseline health score at each cutoff is compared with the accounting benchmarks for five events over the next eight quarters: a bankruptcy petition, severe raw and abnormal cash burn (cumulative free cash flow below -10% of revenue, raw or after subtracting the firm's own trailing run-rate), a fall in interest coverage below $1.5\times$, and a dividend cut. The *health formula* — the twin's scoring rule applied to the opening balance sheet — isolates what the simulation adds to the score's ingredients. The hard version of the bankruptcy test restricts the sample to firms that are already weak: the bottom quintile of Altman's Z″ in the cutoff year or a negative free-cash-flow margin.

Benchmarks. Altman's Z″ (Altman and Hotchkiss, 2006), Ohlson's O-score, Zmijewski's score, interest coverage, net debt to EBITDA, cash to revenue, the current ratio, revenue volatility, the estimated fixed-cost share, size, a static stress test of the demand-freeze scenario, and the two persistence benchmarks: the trailing free-cash-flow margin and capital expenditure over revenue. All are computed from the same vintage as the twin and oriented so that higher means riskier.

5.3 Inference

Bootstrap confidence intervals resample firms (500 resamples in the panel, 2,000 in cross-sections); comparisons between predictors are paired. Out-of-sample logits are ridge-penalized and, for test year Y, trained on cutoffs up to $Y-3$, whose eight-quarter outcomes were known by Y; percentile ranks are computed within each cutoff year. Common shocks within a year enter the regressions as year effects, but with eleven cutoffs the intervals do not account for cross-sectional dependence beyond that and may be somewhat narrow. A pigeonhole bootstrap that resamples firms and years independently gives the twin's incremental correlation against the 16-quarter margin a confidence interval of 0.18–0.26. We run many comparisons; the decision rules above are the only tests we treat as confirmatory, and Appendix Table 19 reports Holm-adjusted p-values for the secondary comparisons.

6 Results: transmission

6.1 Levels

Table 2 and Figure 2 report the 16,960 firm-years (2,679 firms) with eight quarters of realized data. Accuracy in levels falls almost monotonically with model complexity. The 16-quarter margin is the most accurate model, with a median absolute error of 6.9 percentage points of revenue; the

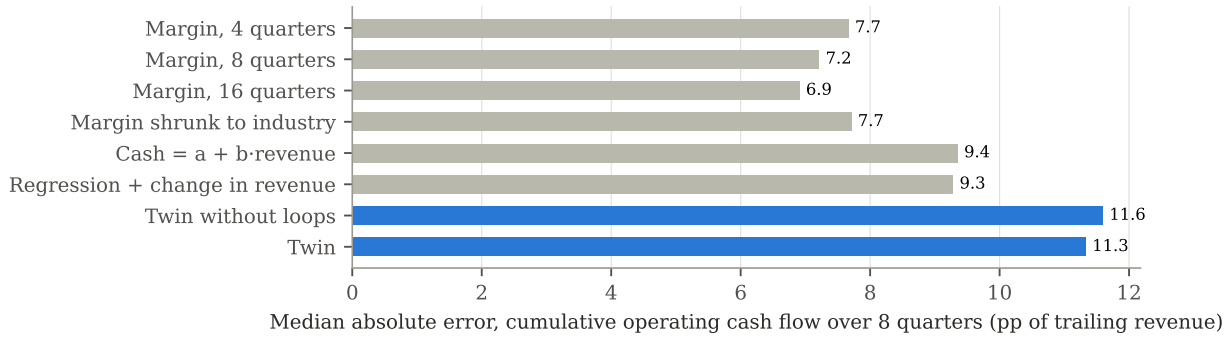


Figure 2: Median absolute error of eight models of cumulative operating cash flow over the eight quarters after each cutoff, given the realized revenue path; 16,960 firm-years, 2014–2024.

Table 2: Transmission: realized revenue given to the twin and to six simple rules

	Operating cash flow					Free cash flow	
	Error (pp of revenue)			Incremental ρ against		Median	Incr. ρ
Model	Median	Mean	ρ	4-quarter margin [95% CI]	16-quarter margin [95% CI]	error	vs. 16q
Margin, 4 quarters	7.7	13.7	0.76		0.21	8.4	0.20
Margin, 8 quarters	7.2	13.1	0.78	0.31 [0.29, 0.33]	0.14	7.8	0.12
Margin, 16 quarters	6.9	12.7	0.80	0.39 [0.38, 0.41]		7.3	
Margin shrunk to industry	7.7	13.5	0.77	0.21 [0.19, 0.22]	0.22	8.2	0.23
Cash = $a + b$·revenue	9.4	15.5	0.75	0.26 [0.25, 0.29]	0.19	10.0	0.18
Regression + change in revenue	9.3	16.0	0.74	0.25 [0.23, 0.27]	0.20	10.1	0.18
Twin without loops	11.6	19.0	0.70	0.31	0.22	12.3	0.24
Twin	11.3	18.7	0.70	0.32 [0.30, 0.34]	0.22 [0.19, 0.24]	12.0	0.24

Outcome: cumulative operating (free) cash flow over the eight quarters after the cutoff, percentage points of trailing revenue. Errors: median and winsorized mean absolute error. ρ: Spearman correlation with the realized outcome. Incremental ρ: correlation of (model − base) with (realized − base); firm-clustered bootstrap CI.

four-quarter margin reaches 7.7, the regressions 9.4 and 9.3, and the twin 11.3. Removing the feedback loops changes little (11.6). The twin's error comes mostly from overreaction: its cost structure, estimated with a prior that 35% of costs are fixed, turns revenue changes into larger cash changes than firms experience (Section 11).

6.2 Information beyond the rules

The first version's statistic, the incremental correlation against the four-quarter margin, is 0.32 for the twin (95% CI 0.30–0.34), close to the 0.30 reported then. Every simple rule scores almost as well, and the 16-quarter margin scores higher (0.39, CI 0.38–0.41): against a noisy base, any less noisy estimate of the persistent margin looks informative. This is the null the referees proposed, and it accounts for all of the original result.

Against the stronger bases the twin still carries information. Its incremental correlation with the residual of the 16-quarter margin is 0.22 (CI 0.19–0.24), with the residual of the shrunk margin 0.32, and with the residual of the regression with timing 0.39. These numbers alone are not decisive — the four-quarter margin itself has an incremental correlation of 0.21 against the 16-quarter margin, because two noisy estimates of the same margin are partly independent — which is why the encompassing regression is the confirmatory test. With the twin, the 16-quarter margin, the shrunk margin and the timing regression in the same regression (Table 3), each keeps a significant coefficient;

Table 3: Forecast encompassing

Cash measure	Specification	4q margin	Twin	16q margin	Shrunk margin	Regression + timing	R^2
Operating cash flow	Departures: twin only		0.40 (16.6)***				0.27
	Departures: simple rules only			0.40 (10.6)***	1.27 (9.1)***	0.33 (10.7)***	0.31
	Departures: all		0.25 (10.6)***	0.31 (8.9)***	0.99 (7.7)***	0.21 (8.3)***	0.39
	Levels: simple rules only	−0.09 (−0.6)		0.34 (8.7)***	0.30 (2.0)*	0.30 (10.1)***	0.67
	Levels: all	−0.16 (−1.1)	0.23 (9.9)***	0.28 (7.6)***	0.35 (2.4)**	0.20 (7.9)***	0.70
Free cash flow	Departures: twin only		0.41 (18.7)***				0.27
	Departures: simple rules only			0.34 (9.1)***	1.35 (13.2)***	0.28 (11.2)***	0.30
	Departures: all		0.26 (12.3)***	0.28 (8.0)***	1.01 (10.5)***	0.16 (8.0)***	0.38
	Levels: simple rules only	−0.34 (−2.8)***		0.29 (7.5)***	0.66 (4.9)***	0.25 (9.9)***	0.60
	Levels: all	−0.37 (−3.3)***	0.24 (11.5)***	0.26 (7.2)***	0.65 (5.1)***	0.15 (7.4)***	0.64

Departures: OLS of (realized − 4-quarter margin rule) on each model's departure from the same rule. Levels: OLS of realized cash flow on the forecasts themselves (Fair and Shiller, 1990). All variables winsorized at the 1st and 99th percentiles; year effects; standard errors clustered by firm; t-statistics in parentheses. *, **, ***: $p < 0.10, 0.05, 0.01$.

the twin's t-statistic is 10.6 for operating and 12.3 for free cash flow. By the pre-specified rule, H2 survives. Because every term in that regression is measured relative to the four-quarter rule, the rule's noise enters the dependent variable and every regressor with the same sign, and the rule's own weight is restricted to one minus the others. The unconstrained form (Fair and Shiller, 1990), which regresses realized cash flow on all five forecasts in levels, gives the same answer: the twin's coefficient is 0.23 with $t = 9.9$ (11.5 for free cash flow). In that regression the four-quarter margin receives a negative weight once the 16-quarter margin is included, the signature of mean reversion.

6.3 How much it is worth

Information that is significant in a regression of 16,000 observations need not be worth much. Table 4 measures it out of sample. Combining the 16-quarter margin with a fraction λ of the twin's departure, with λ chosen each year on earlier years only, lowers the mean absolute error from 12.84 to 12.41 percentage points of revenue, a gain of 0.43 (CI 0.30–0.57), or 3.3% of the error (4.0% for free cash flow). Combining forecasts usually helps even when the second forecast is worse, because their errors are imperfectly correlated (Bates and Granger, 1969; Smith and Wallis, 2009), so the relevant comparison is with the 16-quarter margin combined in the same way with a simple rule. For operating cash flow that comparison removes most of the gain: the margin combined with the cash-revenue regression reaches 12.47 (an improvement of 2.9%), and the twin's combination is better by only 0.05 ($p = 0.332$); against the margin combined with the four-quarter margin it is better by 0.16 ($p = 0.040$). For free cash flow the twin's combination is better than the best combination with a simple rule by 0.22 percentage points (CI 0.06–0.39) and better than each of them ($p = 0.012$ or lower). We had conjectured that this comes from capital expenditure, which the twin cuts under stress and the cash rules do not; combining capex and operating cash flow separately rejects that: capex accounts for only 27% of the gain, and most of it appears only when the two are combined. The estimated weight is small and stable, between 0.10 and 0.15 in every year: the direction of the twin's departures is informative, their size is not. For a company with $10 billion of revenue, the gain over the 16-quarter margin is about $45 million in the predicted cash flow of the next two years, and the gain attributable to the simulation rather than to combining is a fraction of that — relevant at most for a liquidity plan built on a stated revenue scenario, negligible for a credit decision.

Table 4: Out-of-sample combinations

	Operating cash flow		Free cash flow	
Out-of-sample prediction	Mean error	Median error	Mean error	Median error
Margin, 4 quarters	13.79	7.76	15.30	8.40
Margin, 16 quarters	12.84	6.98	14.32	7.39
Regression + revenue change	16.07	9.34	17.95	10.03
Twin alone	18.61	11.42	19.93	11.94
4-quarter margin + λ(twin − rule)	13.01	7.33	14.29	7.90
16-quarter margin + λ(4-quarter margin − rule)	12.56	6.95	13.97	7.35
16-quarter margin + λ(regression − rule)	12.47	6.99	14.01	7.34
16-quarter margin + λ(regression with timing − rule)	12.51	6.99	14.11	7.36
16-quarter margin + λ(twin − rule)	12.41	6.83	13.75	7.17
Difference in mean error (negative: the first is better), firm-clustered CI				
16q margin + twin vs. 16q margin	-0.43 [-0.57, -0.30]***		-0.58 [-0.71, -0.44]***	
16q margin + twin vs. 16q margin + 4q margin	-0.16 [-0.31, -0.01]**		-0.22 [-0.39, -0.06]**	
16q margin + twin vs. 16q margin + regression	-0.05 [-0.13, 0.05]		-0.26 [-0.37, -0.14]***	
16q margin + twin vs. 16q margin + regression with timing	-0.10 [-0.19, 0.00]*		-0.36 [-0.49, -0.24]***	
4q margin + twin vs. 4q margin	-0.78 [-0.91, -0.66]***		-1.01 [-1.16, -0.86]***	
Twin vs. 16q margin	+5.77 [5.16, 6.39]***		+5.60 [4.97, 6.26]***	

Test years 2015–2024; each combination's λ is chosen on earlier years. Errors in percentage points of trailing revenue (mean: winsorized). Differences: firm-clustered bootstrap CI.

Examples. The size of the departures explains the small weight. In 2020, Amazon's revenue over the next eight quarters averaged 53% above its trailing level; the 16-quarter margin implied cumulative operating cash flow of 39% of trailing revenue, the twin 63%, and the company reported 40%. The twin had the direction right for General Electric in 2016 (rule 26%, twin 15%, realized 7%) and wrong for Ford and General Motors in 2020, where revenue changes of −15% and −19% made the twin predict operating cash flow of −0% and −0% while the companies reported 26% and 23% (rules: 19%, 18%); their consolidated cash flows include captive finance operations that a revenue-driven model does not represent. Across firm-years with large departures (more than 8 points) by firms with more than $2 billion of revenue, the twin's departure has the sign of the realized one 58% of the time.

6.4 Where it helps

A referee asked whether the twin is more useful for highly leveraged firms, firms with high fixed costs or large maturities, or in severe revenue contractions. Table 5 gives little support. Its incremental correlation against the timing regression does not rise with leverage (0.43 in the lowest tercile, 0.33 in the highest) and differs little across terciles of fixed costs and maturities. Among the 3,223 firm-years with revenue more than 10% below its trailing level, the twin's incremental correlation against the four-quarter rule (0.20) is lower than the 16-quarter margin's (0.30). Year by year (Figure 3), the twin's statistic against the four-quarter rule is below the 16-quarter margin's in most years, and its statistic against the timing regression is stable around 0.4.

6.5 Why the twin over-reacts, and a revised specification

The twin's errors have a direction. Regressing each eight-quarter total on the revenue change, the twin's operating cash flow responds 0.62 percentage points per point of revenue change against

Table 5: Transmission by firm characteristics and by revenue shock

			Incremental ρ			Median error	
Split by	Group	Firm-years	Twin vs. 4q	16q vs. 4q	Twin vs. regr.	4q margin	Twin
Net debt / EBITDA	low	5,654	0.35	0.43	0.43	7.6	11.2
	mid	5,653	0.31	0.45	0.46	5.8	8.4
	high	5,653	0.34	0.32	0.33	10.4	16.6
Fixed-cost share	low	5,654	0.27	0.38	0.36	6.9	11.5
	mid	5,653	0.33	0.46	0.43	6.7	9.1
	high	5,653	0.35	0.36	0.37	9.9	14.2
Debt due / cash	low	5,654	0.36	0.36	0.39	9.6	13.8
	mid	5,653	0.33	0.39	0.41	7.3	10.7
	high	5,653	0.29	0.45	0.38	6.6	10.0
Revenue over the window	fell >10%	3,223	0.20	0.30	0.39	8.0	12.5
	fell 0-10%	2,854	0.32	0.40	0.25	6.2	7.8
	rose 0-10%	3,845	0.32	0.47	0.35	5.4	8.0
	rose >10%	7,038	0.40	0.41	0.51	10.3	15.5

Terciles of each characteristic at the cutoff; revenue over the window relative to trailing revenue. Incremental ρ as in Table 2; errors in percentage points of trailing revenue.

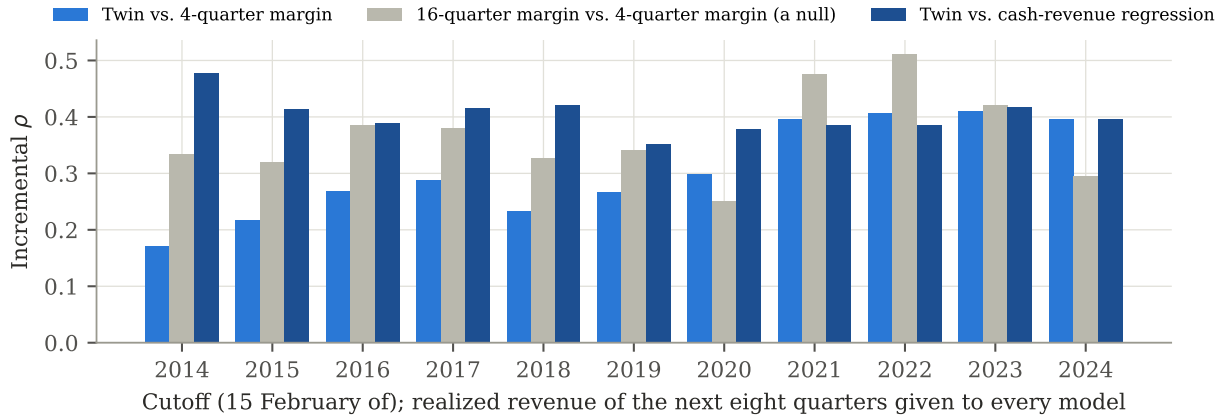


Figure 3: Incremental correlation by cutoff year: the twin and the 16-quarter margin against the 4-quarter margin, and the twin against the cash-revenue regression with timing.

0.27 in reality, a ratio of 2.3; EBITDA shows the same excess (2.1 times), capital expenditure none (0.98). The over-reaction is operating leverage: the cost regression is shrunk toward a prior that 35% of costs are fixed, and firms' costs are more variable than that.

The second-round rule allowed a revised specification only if it was selected without the test year and lowered the twin's error without lowering its bankruptcy AUC by more than 0.01. Choosing the prior each year from a grid (5%–50%, and no prior) on earlier years only, the median error falls from 11.7 to 9.3 (difference 2.4, CI 2.1–2.6) and the bankruptcy AUC moves from 0.839 to 0.838; the revised specification is accepted and is now the software's default. Selecting the prior within each industry gives the same error (9.3). The selected prior is the smallest in the grid in every year, and the unshrunk estimate is close (9.4 against 9.0), so the right amount of shrinkage is closer to none than to the original prior. Even revised, the twin remains less accurate in levels than the 16-quarter margin: the correction removes the over-reaction, not the advantage of persistence.

Table 6: Liquidity events given the realized revenue

			Univariate AUC				Out-of-sample logit AUC		
Event in the next 8 quarters	Firm-years	Rate	Twin	Static runway	Formula	FCF margin	Rule + scores	+ Twin	Δ [95% CI]
Cash falls below half	16,883	29%	0.56	0.53	0.56	0.54	0.56	0.58	+0.021 [0.015, 0.027]***
New borrowing > 5% of revenue	16,960	42%	0.57	0.56	0.53	0.49	0.59	0.60	+0.013 [0.006, 0.019]***
Dividend cut by half	7,941	8%	0.68	0.60	0.69	0.63	0.68	0.69	+0.008 [0.002, 0.015]**
Capex cut by a quarter	16,851	21%	0.65	0.65	0.57	0.60	0.64	0.68	+0.039 [0.031, 0.047]***

Twin: share of simulated paths with the event. Static runway: the trailing FCF margin on realized revenue, minus payouts and unrefinanced maturities (same inputs). Out-of-sample: ridge logits on within-year percentile ranks, trained on cutoffs up to $Y-3$; "Rule + scores" combines the static runway, the health formula and the FCF margin. Δ: firm-clustered bootstrap CI.

Table 7: Feedback loops evaluated on the events they are meant to produce

Event	Full twin	No R1	No R2	No B1	No B2	No loops
Cash falls below half	0.56	0.56***	0.50***	0.57***	0.56	0.50***
New borrowing > 5% of revenue	0.57	0.57	0.57***	0.58***	0.57***	0.57
Dividend cut by half	0.68	0.68	0.68	0.50***	0.68***	0.50***
Capex cut by a quarter	0.65	0.65	0.65	0.62***	0.65	0.62***

AUC of the event probability of the full twin and of the twin with one loop switched off. Stars: paired difference with the full twin.

7 Results: liquidity decisions and feedback loops

Ranking correlations are not what a risk manager uses. Table 6 asks whether, given the revenue that followed, the twin predicts the decisions and events that a liquidity plan is about. Its event probabilities are weak predictors in absolute terms (AUC 0.56–0.68), comparable with the static runway rule and the opening-balance-sheet score, and they add a small amount out of sample to a logit of those rivals: 0.021 in AUC for cash falling below half, 0.013 for new borrowing, 0.008 for dividend cuts and 0.039 for capital-expenditure cuts, all with confidence intervals above zero.

Table 7 evaluates each loop on these outcomes, as a referee suggested. The result is mechanical but informative about the model's structure. Without the market-access cliff (R2), the twin's cash practically never falls below half its opening level, because the treasury always refinances up to its target; the cash-depletion AUC falls from 0.56 to 0.50. Without cash conservation (B1), dividends are never cut and capital expenditure is cut less often (AUC 0.62 against 0.65). The spread spiral (R1) and delayed restructuring (B2) change no outcome. Two of the four loops are therefore necessary for the model to produce the liquidity events at all; the other two are inert over an eight-quarter horizon. None of them changes the ranking of firms (Section 8).

The loops also matter for timing. Ranking firms by the twin's probability of a cash depletion within eight quarters gives a concordance index for the time to the event of 0.55, against 0.50 without loops (difference 0.05, CI 0.04–0.07) and 0.52 for the static runway; for new borrowing the loops change nothing (0.56 against 0.56). The results hold at other thresholds for each event (Appendix Table 17). By the pre-specified rule — a gain from the loops in at least two of the four events and in the timing of at least one — H3 is accepted.

Table 8: Bankruptcy petitions (8-K Item 1.03), all non-financial filers

	Panel 2014–2024, within 8 quarters		20 Feb 2020, to end-2021	
	All (419 events)	Already weak (342)	All (55)	Already weak (40)
Twin: health score (panel: baseline; 2020: stress)	0.85 [0.83, 0.88]	0.77 [0.74, 0.80]	0.85 [0.80, 0.89]	0.75 [0.68, 0.81]
Twin: breach probability	0.84 [0.81, 0.86]	0.77 [0.74, 0.80]	0.79 [0.73, 0.83]	0.68 [0.61, 0.74]
Health formula, no simulation	0.87 [0.85, 0.88]	0.77 [0.74, 0.79]	0.85 [0.80, 0.89]	0.74 [0.67, 0.80]
Altman Z″	0.81 [0.79, 0.84]	0.66 [0.63, 0.69]	0.80 [0.75, 0.85]	0.62 [0.55, 0.69]
Ohlson O-score	0.86 [0.83, 0.88]	0.77 [0.74, 0.80]	0.80 [0.72, 0.87]	0.73 [0.64, 0.82]
Zmijewski score	0.82 [0.80, 0.85]	0.76 [0.73, 0.78]	0.78 [0.72, 0.85]	0.74 [0.66, 0.82]
Interest coverage	0.79 [0.77, 0.82]	0.65 [0.62, 0.67]	0.77 [0.71, 0.82]	0.61 [0.54, 0.68]
Net debt / EBITDA	0.78 [0.76, 0.81]	0.66 [0.64, 0.69]	0.74 [0.67, 0.80]	0.62 [0.54, 0.69]
Cash / revenue	0.60 [0.56, 0.63]	0.58 [0.54, 0.62]	0.64 [0.57, 0.71]	0.63 [0.56, 0.71]
Current ratio	0.68 [0.65, 0.71]	0.61 [0.58, 0.65]	0.72 [0.65, 0.78]	0.63 [0.56, 0.71]
Static stress test	0.78 [0.75, 0.80]	0.71 [0.68, 0.74]	0.76 [0.70, 0.82]	0.66 [0.58, 0.74]
Revenue volatility	0.67 [0.63, 0.70]	0.60 [0.56, 0.64]	0.63 [0.55, 0.72]	0.56 [0.45, 0.66]
Fixed-cost share	0.67 [0.64, 0.71]	0.63 [0.59, 0.67]	0.68 [0.60, 0.75]	0.61 [0.51, 0.71]
Size (log revenue)	0.63 [0.60, 0.66]	0.52 [0.48, 0.56]	0.56 [0.50, 0.63]	0.41 [0.34, 0.49]
Trailing FCF margin	0.78 [0.76, 0.81]	0.58 [0.54, 0.62]	0.77 [0.71, 0.83]	0.52 [0.42, 0.62]
Capex / revenue	0.50 [0.46, 0.55]	0.48 [0.43, 0.53]	0.50 [0.39, 0.61]	0.48 [0.38, 0.59]

AUC with firm-clustered 95% CI. Panel: baseline health score at each cutoff; 2020: stress health score on 20 February 2020. Already weak: bottom quintile of Altman's Z″ in the cutoff year or negative trailing FCF margin.

Table 9: Distress events over the next eight quarters, 2014–2024

			Univariate AUC			Out-of-sample logit AUC		
Event (next 8 quarters)	Firm-years	Events	Twin	Formula	Best benchmark	Benchmarks	+ Twin	Δ [95% CI]
Severe cash burn	19,005	2847	0.72	0.71	0.82 (trailing FCF margin)	0.87	0.87	+0.002 [0.001, 0.003]***
Severe abnormal burn	16,025	3727	0.52	0.51	0.62 (revenue volatility)	0.68	0.68	−0.004 [−0.007, −0.001]**
Coverage collapse	9,107	1956	0.58	0.59	0.66 (interest coverage)	0.69	0.70	+0.005 [0.002, 0.008]***
Dividend cut	8,781	767	0.69	0.69	0.69 (Ohlson O-score)	0.74	0.75	+0.002 [−0.004, 0.008]
Bankruptcy petition	21,337	419	0.85	0.87	0.86 (Ohlson O-score)	0.87	0.87	+0.001 [−0.009, 0.009]

Twin: baseline health score. Out-of-sample logits trained for each test year on cutoffs at least three years earlier; "Benchmarks" contains all thirteen benchmarks and the formula, "+ Twin" adds the twin's baseline health, breach probability and recession-scenario health. Δ: firm-clustered CI.

8 Results: ranking firms

Table 8 reports the bankruptcy test on the census. Across 21,337 firm-years with 419 petitions within eight quarters, the twin's baseline score has an AUC of 0.85, against 0.87 for its own formula on the opening balance sheet (difference -0.012, $p = 0.112$), 0.86 for Ohlson's O-score and 0.81 for Altman's Z″. Size, which separated the 2020 bankruptcies from S&P members in the first version, now reaches only 0.63. The hard task is the one a referee emphasized: 82% of the petitions come from the 6,993 firm-years that were already weak. Among them the twin reaches 0.77 and the formula 0.77 ($p = 0.580$ for the difference); Altman's Z″ falls to 0.66. The same pattern holds for the 55 petitions after 20 February 2020 (twin 0.85, formula 0.85). The twin's score is a good distress score; the simulation behind it adds nothing to the ranking.

Table 9 gives the other events. For severe cash burn the twin reaches 0.72, below the trailing free-cash-flow margin (0.82), and adds +0.002 out of sample to a logit of all benchmarks and the formula; for abnormal burn, the deterioration a stress test claims to anticipate, it has no skill (0.52). Fundamentals-based fragility, the twin's included, largely measures persistent cash-flow habits.

Table 10: 2020–21 cash burn, predictors frozen on 20 February 2020

	All filers ($n = 1944$)		S&P 500 on 20 Feb 2020 ($n = 382$)	
	Raw burn	Abnormal burn	Raw burn	Abnormal burn
Twin: stress health score	0.32	−0.04	0.45	0.04
Twin: baseline health score	0.37	0.04	0.38	0.07
Health formula, no simulation	0.34	0.00	0.39	0.04
Altman Z″	0.30	0.04	0.36	0.10
Ohlson O-score	0.28	−0.03	0.25	0.06
Zmijewski score	0.19	−0.01	0.20	0.07
Interest coverage	0.26	0.01	0.29	0.13
Net debt / EBITDA	0.26	0.00	0.28	0.04
Cash / revenue	0.03	−0.17	0.31	−0.11
Current ratio	0.09	−0.03	0.27	0.03
Static stress test	0.20	−0.09	0.33	−0.05
Revenue volatility	0.18	0.11	0.13	0.20
Fixed-cost share	0.14	0.10	0.02	0.13
Size (log revenue)	0.18	0.07	−0.14	0.05
Trailing FCF margin	0.57	−0.13	0.70	0.01
Capex / revenue	0.10	0.05	0.20	0.18

Spearman correlation between each risk-oriented predictor and realized burn severity. Raw burn: worst cumulative FCF in 2020Q1–2021Q2 as a share of trailing revenue; abnormal burn: the same after subtracting each company's pre-2020 run-rate.

9 Downturns: 2020 and two others

The first version built its ranking evidence on the COVID-19 shock; a referee rightly noted that an exogenous pandemic is not something an early-warning system can be expected to foresee. It is now one cutoff among eleven, and we use it for a question the uniform scenario could not answer: is the failure to rank the shock a failure of the model or a lack of information about exposure? Table 10 repeats the cross-section for the 1944 filers alive on 20 February 2020 and for the 382 members of the S&P 500 on that date. The pattern of the first version holds without survivorship: the twin's stress score ranks raw cash burn (0.32 in the full sample, 0.45 in the S&P), the free-cash-flow margin ranks it better (0.57, 0.70), and nothing ranks abnormal burn.

Table 12 gives each firm the median realized revenue shock of its industry peers (leave-one-out, 2-digit SIC; median depth 17%) instead of the uniform freeze. Read at that depth, the twin ranks raw burn better (0.40 against 0.32). For abnormal burn, the industry shock alone has a modest correlation (0.13), and in a rank regression that also contains the FCF margin, the twin's stress score at the industry shock is significant ($t = 5.3$). Given the exposure, fragility does predict who suffered more than usual; what filings cannot supply is the exposure.

The same design applies to any downturn in the panel. Table 11 gives each firm its industry's realized revenue shock over the eight quarters after the February 2015 cutoff (the commodity bust), the February 2020 cutoff and the February 2022 cutoff (the rate shock), reads the twin's score under a generic demand shock of that depth, and regresses abnormal burn on the shock, the twin and the free-cash-flow margin. The twin's t-statistic is 3.3, 5.9 and 4.0; by the pre-specified rule (above 2 in 2015 and 2022) the finding generalizes. A filing-based twin cannot say who a downturn will hit, but once the hit is known it says who will bleed most.

Table 11: Fragility given exposure in three downturns

Episode	Firms	Median shock	Burn	Rank regression t: Industry shock	Twin at shock	FCF margin
2015–16 commodity bust (cutoff Feb 2015)	1,844	9%	raw	1.8	4.6	20.3
			abnormal	2.5	3.3	−9.3
2020 pandemic (Feb 2020)	1,846	18%	raw	0.6	7.1	19.1
			abnormal	2.9	5.9	−7.2
2022 rate shock (Feb 2022)	1,899	8%	raw	4.2	5.2	23.7
			abnormal	3.1	4.0	−8.7

Rank regression of burn severity over the eight quarters after the cutoff on the percentile ranks of the industry's leave-one-out realized revenue shock, the twin's stress score at that shock and the trailing FCF margin; HC3 t-statistics. The shock uses realized information: this is a decomposition, not a forecast.

Table 12: Fragility and exposure in 2020

Outcome	Spearman ρ with burn: Twin, uniform freeze	Industry shock alone	Twin at industry shock	FCF margin	Rank regression, coefficient (t): Industry shock	Twin at shock	FCF margin
Raw burn	0.32	0.13	0.40	0.57	0.08 (3.6)	0.17 (6.4)	0.49 (18.8)
Abnormal burn	−0.04	0.13	0.07	−0.13	0.12 (4.9)	0.15 (5.3)	−0.21 (−7.1)

Industry shock: leave-one-out median of peers' worst year-on-year revenue change in 2020–21 (2-digit SIC with at least five peers, else Fama–French industry); it uses realized information and is a decomposition, not a forecast. Rank regression: OLS of the percentile rank of burn severity on percentile ranks, HC3 t-statistics.

10 Forecasts and calibration

For S&P 500 members (today's and those of 2020), the twin's four-quarter predictive distributions of revenue improve on a seasonal random walk by 26% and on exponential smoothing by 19% in scaled CRPS ($t = -5.83$ and -3.93; Table 13); for cash both benchmarks are better (−29% and −37%). Blending the twin into the random walk for cash, with the weight chosen on earlier origins, meets the pre-specified rule only trivially: the skill is 0.3% (CI 0.1%–0.4%) because the selected weight falls from 0.15 to 0.00 after 2018, that is, the best cash forecast is the random walk.

Raw 90% intervals are too narrow: they cover 70% of revenue outcomes and 36% of cash outcomes. Online conformal rescaling, using only errors whose target quarter ended at least 75 days before the forecast origin (by then most quarters, including fourth quarters reported in 10-Ks, have been filed), restores coverage to 91% and 91% on average (Table 14, Figure 4). The referee's question was whether this holds where it matters. It holds across horizons and firm sizes, but not in the stress year: for quarters from 2020Q2 to 2021Q1, conformal revenue intervals cover only 77%. The recalibration learns the average error of quiet years and cannot anticipate a regime break. For cash, the raw intervals do not widen for smaller firms ($t = -0.6$ for size in a regression of interval width) although their errors are larger ($t = -9.0$): the uncertainty model is too homogeneous across firms.

Two remedies were tested under a pre-specified target of 85–95% coverage in every group (Appendix Table 18). Rescaling by size and volatility tercile, with factors learned from reported errors, gives 90% coverage for cash and 89% for revenue; it meets the target for cash and misses it for revenue. Adaptive conformal inference (Gibbs and Candès, 2021) reaches 91% and 90% and misses the target for revenue. Neither fixes the regime break: for the 2020Q2–2021Q1 quarters revenue coverage is 78% with group scaling and 77% with adaptive conformal, which can only react

Table 13: Four-quarter probabilistic forecasts

			vs. seasonal random walk			vs. ETS		
	Pairs	Firms	Skill	Better	DM t	Skill	Better	DM t
Revenue	18,792	469	26%	68%	−5.83	19%	58%	−3.93
Cash and short-term investments	19,144	474	−29%	44%	5.77	−37%	41%	7.10

Skill: $1 - \overline{\text{CRPS}}_{\text{twin}}/\overline{\text{CRPS}}_{\text{benchmark}}$, CRPS scaled by mean quarterly revenue. Better: share of pairs where the twin's CRPS is lower. DM t: Diebold–Mariano statistic clustered by firm and origin (negative: twin better).

Table 14: Calibration of 90% predictive intervals

	90% coverage		Conformal 90% coverage, by				Below	Mean
	Raw	Conformal	Other targets	2020Q2–2021Q1	$h = 1$ / 4	Small / large firms	5% quantile	PIT
Revenue	70%	91%	92%	77%	89% / 92%	91% / 90%	17%	0.48
Cash	36%	91%	91%	90%	93% / 89%	89% / 93%	38%	0.42

after the errors are reported.

11 Sensitivity and hindsight

Several parameters are judgement calls rather than estimates: the credit line assumed when none is tagged, the prior on the fixed-cost share and its weight, the shrinkage of trend growth and cyclicality, the coverage thresholds of the market-access cliff, the speed of restructuring, and the rating-spread table. Table 15 varies each in turn. The incremental correlation against the four-quarter rule ranges from 0.30 to 0.35, and the bankruptcy AUC from 0.83 to 0.84. The one assumption that matters is the fixed-cost prior: with a prior of 20% instead of 35% the twin's median error falls to 9.9 points, with 50% it rises to 13.3. The data prefer less operating leverage than the model assumes, which is the overreaction seen in Section 6. Varying all eight assumptions jointly over wide ranges (a Saltelli design of 640 points, each recalibrating the twin for 317 random firm-years; Appendix Table 20), the median error ranges from 8.6 to 18.0 and the incremental correlation against the four-quarter margin from 0.15 to 0.25; the fixed-cost prior alone has a total-effect index of about 1.00 for the error's variance, against 0.13 for all other assumptions together.We report the specified model and do not retune it on the test sample.

Hindsight enters in three places, each now addressed or measured. GDP enters as the vintage published on each cutoff. Rating spreads come from the version of Damodaran's table that was published before each cutoff, recovered from the Internet Archive (for 2014 and 2021, whose archived copies postdate February, the previous year's table); using the 2026 table instead, as the software does, changes nothing (Table 15). The model's structure and priors were designed in 2026 with knowledge of the 2014–2024 data, which no backtest can undo; the decision rule for H2 was fixed before the full-sample test, and the weekly rebuild of the public twin now accumulates genuinely prospective predictions.

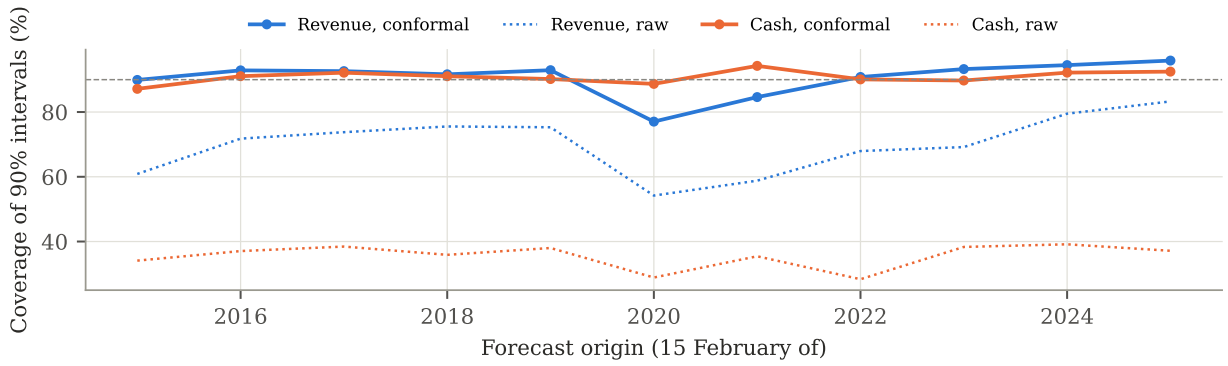


Figure 4: Coverage of raw and conformally rescaled 90% intervals by forecast origin.

Table 15: Sensitivity to assumed parameters

	Median error	Incremental ρ against		Baseline score AUC	
Assumption	(pp of revenue)	4q margin	regression	Bankruptcy	Severe burn
As specified	11.3	0.32	0.39	0.84	0.73
Credit line 4% of revenue (untagged)	11.3	0.32	0.39	0.84	0.73
Credit line 15% of revenue (untagged)	11.3	0.32	0.39	0.84	0.74
Fixed-cost prior 20%	9.9	0.34	0.43	0.84	0.75
Fixed-cost prior 50%	13.3	0.30	0.34	0.83	0.72
Prior weight 0.5 (data dominate)	10.0	0.34	0.41	0.84	0.75
Prior weight 8 (prior dominates)	12.4	0.31	0.37	0.84	0.73
No shrinkage of growth and cyclicality	11.3	0.32	0.39	0.84	0.73
Market access cut below 2.5× / 1.5× coverage	11.3	0.32	0.39	0.84	0.73
Market access cut below 1.0× / 0.5×, milder cuts	11.3	0.32	0.39	0.84	0.73
Faster, deeper restructuring	11.1	0.33	0.40	0.84	0.73
January 2026 rating table (as in the product)	11.4	0.32	0.39	0.84	0.73
2026 table rescaled by the cutoff's Baa spread	11.5	0.32	0.38	0.84	0.73

12 Discussion

The hypotheses. H1 is rejected: the twin's stress and baseline scores do not rank distress better than the trailing free-cash-flow margin or than its own scoring formula without simulation. H2 is accepted by the pre-specified rule, with a small economic magnitude. H3 is accepted by its pre-specified rule: two of the four loops are necessary for the model to produce the liquidity events, improve their prediction and their timing slightly; the other two have no measurable effect.

Statistical, predictive and decision value. The three are different, and the evidence separates them. The twin's transmission mechanics are statistically informative beyond persistence (a t-statistic near 10 in the encompassing regressions). Their predictive value is small: added with a weight of 0.10–0.15 to the best simple rule they lower its error by a few percent, most of which a simple rule added the same way also delivers, except for free cash flow. Their decision value lies where a simple rule has no answer at all: the timing of a breach when maturities and credit lines interact, the amount of new borrowing a stated scenario requires, whether a dividend survives. On those questions the twin is a structured calculator whose inputs are estimated and audited from filings, and its advantage over a static runway calculation is small but positive.

What the twin is for. A filing-based twin is not an early-warning system: its ranking of firms is that of its own scoring formula. It is a tool for conditional analysis of a specified shock, and its output should be shrunk toward persistence — in practice, a cash forecast should start from the long-window margin and adjust a fraction of the way toward the simulation.

External validity. The universe is US non-financial SEC filers with at least $50 million of revenue. The pipeline depends on standardized, tagged, point-in-time disclosure, which exists in a comparable form only under SEC XBRL and, since 2020–21, the European Single Electronic Format for annual reports of listed companies (without quarterly data). Among the 1,275 firm-years of firms with $10–50 million of revenue, the pattern is the same: the twin's bankruptcy AUC is 0.72 against 0.72 for its formula (21 petitions), and its median cash-flow error is 40.6 against 32.3 for the 16-quarter margin. Private firms report less often and less consistently, which affects calibration more than structure: the stock-and-flow core and the accounting identity carry over, but parameters estimated from 16 quarters would have to come from fewer annual observations, with more weight on priors. Banks and insurers need a different balance-sheet model. Market-based inputs, which structural credit models use, are a natural extension for listed firms.

Limitations. The transmission test uses realized revenue, so it cannot separate errors in the model's mechanics from noise in reported cash flow. Bankruptcy is observed only through 8-K filings; firms that stop filing before petitioning, or that are acquired in distress, are not counted. Outcomes for firms that disappear are missing, so the transmission and burn tests condition on survival for eight quarters. Some XBRL items are tagged idiosyncratically.

13 Conclusion

We built a transparent, automatically calibrated simulation model of the firm and asked what it knows that simple rules do not. Most of what it appeared to know was persistence, and the test that first suggested otherwise measured mean reversion. What remains is small and specific: given a revenue path, the direction of the simulation's departures from a long-window cash-flow margin is informative in an encompassing regression, and adding a small fraction of them to the margin improves free-cash-flow predictions out of sample by more than any simple rule does. The simulation does not help to rank firms by distress. These answers come from a design that any simulation model of the firm can be held to — a ladder of simple rivals, an encompassing test and an out-of-sample combination, on a survivorship-free panel with a census of defaults — and the data, code and tests are public so that it can be.

Data and code availability

All code, the point-in-time pipeline, the EDGAR universe and bankruptcy census, the research experiments and the scripts that generate every table, figure and number in this paper are at `https://github.com/SpiliosDimakopoulos/fintwin` under the MIT licence and archived on Zenodo (doi:10.5281/zenodo.23434606). The experiments run on free GitHub-hosted runners (`.github/workflows/research.yml`). An interactive version of the twin is at `https://`

spiliosdimakopoulos.github.io/fintwin/. All inputs are public: SEC EDGAR, FRED and the Philadelphia Fed Real-Time Data Set.

Use of AI tools

Software development, data analysis and drafting were assisted by an AI coding assistant (Claude, Anthropic). The author designed the study, reviewed all code, results and text, and takes full responsibility for the content.

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A Additional material

Table 16: Calibrated parameters

Parameter	Estimate	Window
Seasonal factors S_q	median ratio to centred 4-quarter moving average	last 32 quarters
Trend growth μ	median log growth, shrunk 50% toward 1.2%/quarter	last 12 quarters
AR coefficient ϕ, volatility σ_g	OLS; median absolute deviation of residuals	last 32 quarters
Tail index	Student-t maximum likelihood on standardized residuals	growth residuals
Cyclicality β	regression of revenue on real-time nominal GDP growth, shrunk 50% toward 1.5	last 40 quarters
Recession entry and exit	NBER recessions since 1947	FRED
Fixed costs F, variable ratio v	penalized regression of cash costs on revenue, prior 35% fixed	last 16 quarters
Margin shock m_t	AR(1) of cost-regression residuals; correlation with demand	last 16 quarters
DSO, DIO, DPO, deposit days	averages of balance to revenue or costs	last 4 quarters
Depreciation, capex, SBC ratios	medians	last 8 quarters
Coupon	interest expense over debt	last 4 quarters
Rollover share	two-year debt trend relative to maturities	last 8 quarters
Discretionary financing volatility	robust s.d. of quarterly debt changes	last 20 quarters
Maturity ladder	XBRL maturity schedule, else six-year average life	latest 10-K
Credit facility	XBRL facility size, else 8% of revenue (assumption)	latest filing
Minimum cash	half the 10th-percentile cash balance	last 12 quarters
Short-rate dynamics	Vasicek fit to the 3-month Treasury bill	since 2000

Table 17: Liquidity events at other thresholds (given realized revenue)

Event	Threshold	Rate	Twin AUC	No loops AUC
Cash falls below share of opening	30%	14%	0.56	0.50
Cash falls below share of opening	50%	29%	0.56	0.50
Cash falls below share of opening	70%	50%	0.58	0.59
New borrowing above share of revenue	3%	48%	0.58	0.58
New borrowing above share of revenue	5%	42%	0.57	0.57
New borrowing above share of revenue	10%	32%	0.57	0.57
Dividend cut by at least	25%	12%	0.69	0.50
Dividend cut by at least	50%	8%	0.68	0.50
Capex cut by at least	25%	21%	0.65	0.62
Capex cut by at least	50%	8%	0.64	0.60

Table 18: Coverage of 90% intervals with group-conditional and adaptive conformal calibration

Variable	Split	Group	Group-conditional	Adaptive (ACI)
Revenue	Target quarters	2020Q2-2021Q1	78%	77%
Revenue	Target quarters	other	91%	92%
Revenue	Size tercile	low	89%	91%
Revenue	Size tercile	mid	90%	89%
Revenue	Size tercile	high	90%	90%
Revenue	Volatility tercile	low	89%	87%
Revenue	Volatility tercile	mid	90%	91%
Revenue	Volatility tercile	high	89%	92%
Cash	Target quarters	2020Q2-2021Q1	89%	90%
Cash	Target quarters	other	90%	91%
Cash	Size tercile	low	91%	89%
Cash	Size tercile	mid	90%	91%
Cash	Size tercile	high	90%	93%
Cash	Volatility tercile	low	90%	89%
Cash	Volatility tercile	mid	91%	92%
Cash	Volatility tercile	high	90%	91%

Table 19: Holm-adjusted p-values of the secondary comparisons

Secondary comparison	Holm-adjusted p
Liquidity: capex cut, twin added	<0.001
Liquidity: cash halves, twin added	<0.001
Liquidity: new borrowing, twin added	<0.001
Combination: FCF twin vs. 16q + regression	<0.001
Combination: FCF twin vs. 16q + regression with timing	<0.001
Distress: severe burn, twin beyond formula	<0.001
Distress: coverage collapse, twin beyond formula	0.044
Liquidity: dividend cut, twin added	0.120
Combination: FCF twin vs. 16q + 4q	0.120
Distress: abnormal burn, twin beyond formula	0.160
Combination: OCF twin vs. 16q + 4q	0.280
Combination: OCF twin vs. 16q + regression with timing	0.336
Bankruptcy AUC: twin minus formula, all firms	0.560
Combination: OCF twin vs. 16q + regression	1.000
Bankruptcy AUC: twin minus formula, weak firms	1.000
Distress: dividend cut, twin beyond formula	1.000
Distress: bankruptcy, twin beyond formula	1.000

B Implementation

The Python reference engine (NumPy) performs calibration, experiments and exports. A line-for-line TypeScript port runs in a Web Worker, so the public dashboard needs no server. Golden runs exported from Python with identical pre-drawn shocks must agree to 10^{-6}; the balance-sheet identity is checked by property-based tests; the data pipeline, including 52/53-week fiscal calendars, is tested end to end on synthetic XBRL; and a synthetic test checks that the transmission statistics reject a predictor that is merely smoother than the base rule. The research sample (5,220 companies) is run in 16 shards on free GitHub-hosted runners.

Table 20: Global (Sobol) sensitivity of the transmission result to the assumed parameters

		Median error		Incremental ρ	
Assumption	Range	S_1	S_T [95% CI]	S_1	S_T
Fixed-cost prior	0.02–0.6	1.00	1.00 [0.79, 1.00]	0.60	0.85
Weight of the prior	0–10	0.00	0.13 [0.06, 0.23]	0.11	0.11
Credit line when untagged	0.02–0.2	0.00	0.00 [0.00, 0.00]	0.00	0.00
Weight on own trend growth	0–1	0.00	0.00 [0.00, 0.00]	0.00	0.00
Weight on own GDP beta	0–1	0.00	0.00 [0.00, 0.00]	0.00	0.00
Coverage below which access is cut	1–3	0.00	0.00 [0.00, 0.00]	0.00	0.00
Restructuring speed	0–0.1	0.00	0.00 [0.00, 0.01]	0.00	0.00
Restructuring floor	0.7–1	0.00	0.00 [0.00, 0.00]	0.00	0.00

First-order (S_1) and total (S_T) Sobol indices of the median absolute error and of the incremental correlation against the four-quarter margin, over a joint uniform design of 64 base samples; bootstrap CI over the base samples. With so few base samples the estimators are noisy: indices are truncated to $[0, 1]$ and first-order indices to the total index.